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CLINICAL RESEARCH
Ablation for atrial fibrillation

Individually tailored vs. standardized substrate modification during radiofrequency catheter ablation for atrial fibrillation: a randomized study

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Aims This randomized single-centre study sought to compare the efficacy and safety of pulmonary vein isolation (PVI) plus voltage-guided ablation vs. PVI with or without linear ablation depending on the type of atrial fibrillation (AF).

Methods and results Overall, 124 ablation-naive patients with paroxysmal or persistent AF were randomized to PVI with (persistent AF) or without (paroxysmal AF) additional linear ablation (control group) vs. PVI plus ablation of low-voltage areas (LVAs) irrespective of AF type. Bipolar voltage mapping was performed during stable sinus rhythm. An LVA consisted of ≥3 adjacent mapping points that each had a peak-to-peak amplitude ≤0.5 mV. After a mean follow-up of 12 ± 3 months, significantly more patients in the LVA ablation group were free from atrial arrhythmia recurrence >30 s off antiarrhythmic drugs (AADs) after a single procedure (primary endpoint) compared with control group patients [40/59 (68%) vs. 25/59 (42%), log-rank $P = 0.003$]. Arrhythmia-free survival on or off AADs was found in 33/59 control group patients (56%) and in 41/59 LVA ablation group patients (70%) (adjusted log-rank $P = 0.10$). During the 7 day Holter monitoring period at 12 months, significantly more patients in the LVA ablation group were free from arrhythmia recurrence on or off AADs [45/50 (90%) vs. 33/46 (72%), $P = 0.04$]. No between-group differences were observed regarding procedure duration, fluoroscopy time, and major complications.

Conclusion In this single-centre study, individually tailored substrate modification guided by voltage mapping was associated with a significantly higher arrhythmia-free survival rate compared with a conventional approach applying linear ablation according to AF type.

Keywords Atrial fibrillation • Catheter ablation • Fibrosis • Voltage mapping • Substrate modification

Introduction

Radiofrequency (RF) catheter ablation is an established therapy for symptomatic atrial fibrillation (AF).[1,2] Whilst pulmonary vein isolation (PVI) is performed as a stand-alone procedure in the majority of patients with paroxysmal AF, additive substrate modification is frequently applied in persistent forms of AF to improve ablation outcome.[3,4] Presently, however, there is no uniform concept for the definition of

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What's new?

- Low-voltage areas (LVAs) detected during left atrial electroanatomic mapping may represent potentially proarrhythmogenic structural alterations of the atrial myocardium (e.g. fibrosis) and may thus serve as ablation targets for individualized substrate modification during catheter ablation for atrial fibrillation (AF).
- Data concerning the clinical benefit of voltage-guided substrate modification are scarce.
- This is the first randomized trial to investigate the role of voltage-guided substrate modification in patients with paroxysmal or persistent AF.
- After a mean follow-up of 12 ± 3 months, pulmonary vein isolation (PVI) plus ablation of LVAs irrespective of AF type was associated with a significantly higher single-procedure arrhythmia-free survival rate off antiarrhythmic drugs compared with a conventional approach consisting of PVI with (persistent AF) or without (paroxysmal AF) additional linear ablation.

atrial substrate. Left atrial (LA) structural alterations, especially fibrosis, are major determinants of AF pathogenesis and predict procedural failure after ablation.[5–8] Low-voltage areas (LVAs) detected by LA electroanatomic mapping correlate well with structural alterations on delayed-enhancement magnetic resonance imaging (DE MRI) and may thus serve as a suitable target for adjunctive substrate modification.[9] Accordingly, we hypothesized that a patient-specific ablation approach consisting of PVI plus individually tailored substrate modification guided by endocardial voltage mapping (EVM) would be associated with a better rhythm outcome compared with a conventional strategy consisting of PVI with or without additional application of anatomically defined linear lesions depending on the type of AF.

Methods

Study design

This prospective, randomized, single-centre trial was approved by the local ethics committee. All patients provided written informed consent before enrolment. Patients with symptomatic paroxysmal, persistent, or long-standing persistent AF were eligible for study participation if they were ≥18 years; had ≥2 documented AF episodes within the preceding 6 months; and had failed or were intolerant to ≥1 Class I or Class III antiarrhythmic drug (AAD), or refused AADs. Key exclusion criteria were a previous AF ablation, an LA diameter >60 mm, intra-atrial thrombi, significant coronary artery or valvular heart disease, and a contraindication for anticoagulant therapy. 'Paroxysmal' AF was defined as recurrent AF that was self-terminating within 7 days or that was cardioverted either electrically or pharmacologically within 48 h after AF onset. Atrial fibrillation was classified as 'persistent' if episodes lasted >7 days or if cardioversion had been performed >48 h after AF onset. These classifications were based on the observation that paroxysmal AF episodes frequently terminate spontaneously within 48 h.[10] Long-standing persistent AF was defined as continuous AF >12 months.[1,2] Patients were randomly assigned to conventional (control group) or individually tailored (LVA ablation group) substrate modification in a 1:1 fashion using sealed envelopes for block randomization (see Supplementary material online, *Figure S1*). Patients were not actively informed about the treatment allocation. Blinding of patients, however, could not be fully ensured, because patients had access to the ablation report incorporated in the discharge letter.

The primary endpoint was freedom from any atrial arrhythmia recurrence off Class I/III AADs during a 12-month follow-up period after a single ablation procedure. Arrhythmia recurrence was defined as any episode of AF, atrial flutter (AFLU), or atrial tachycardia (AT) >30 s documented beyond a 90 day post-ablation blanking period (BP).[1,2] As the mechanism of a regular supraventricular tachycardia (i.e. macro-reentrant or focal) cannot be unequivocally established from surface electrocardiographic (ECG) recordings 'AFLU' and 'AT' were summarized as 'organized ATs'. Secondary efficacy endpoints included single-procedure arrhythmia-free survival on or off AADs during the follow-up period, and single-procedure freedom from arrhythmia recurrence during the 7 day Holter ECG recording periods at 6 and 12 months. Safety endpoints encompassed all procedure-related adverse events.

Mapping and ablation

Antiarrhythmic drugs were discontinued ≥24 h prior to ablation. A pre-procedural transoesophageal echocardiography was performed to exclude LA thrombi. Trans-septal access to the left atrium was obtained with standard techniques. Repetitive intravenous heparin boli were administered to maintain an activated clotting time of 300 to 350 s. Catheter navigation and mapping were performed with the use of an electroanatomic mapping system (Ensite Velocity, St. Jude Medical Inc., St. Paul, MN, USA; or CARTO, Biosense Webster, Diamond Bar, CA, USA).[11] An intra-oesophageal temperature probe was used for continuous real-time monitoring of the intraluminal temperature during RF delivery (Sensitherm, St Jude Medical). Radiofrequency ablation was performed with externally irrigated-tip catheters (Navistar Thermocool, Biosense Webster; or Therapy CoolPath, St. Jude Medical). Standard generator settings consisted of a maximum tip temperature of 48 °C, a power of 35 W, and an irrigation rate of 30 mL/min. Radiofrequency power was empirically reduced to 25 W at the LA posterior wall and subsequently adjusted according to intra-oesophageal temperature.

In both groups, wide area circumferential PVI was performed as the initial step with bidirectional conduction block serving as procedural endpoint (see Supplementary material online, *Figure S1*). If patients presented with AF at the beginning of the procedure or if AF occurred spontaneously during the procedure, sinus rhythm was restored by electrical cardioversion either before ablation or after encircling of the pulmonary veins (PVs). After PVI, a detailed LA bipolar voltage map was created as described previously.[12] In brief, approximately 120 evenly distributed mapping points were systematically acquired during stable sinus rhythm with a multipolar circular mapping catheter (Inquiry Optima, St. Jude Medical; or Lasso, Biosense Webster) using an interpolation threshold of 10 mm. Subsequently, the ablation catheter was used to map sites that were not readily accessible with the circular mapping catheter and to precisely define the extent of an LVA. Appropriate endocardial wall contact was assessed by electrogram stability, fluoroscopy, tactile feedback, and 'distance-to-surface' measurements on the 3D mapping system. Bipolar electrograms were filtered at 30 and 500 Hz. Local peak-to-peak electrogram amplitudes were defined as follows[9,12–14]: Bipolar voltage amplitudes >0.5 mV were considered 'normal', whereas amplitudes ≤0.5 mV were defined as 'diseased' (0.2–0.5 mV) or 'scar' (<0.2 mV). An LVA was defined by ≥3 adjacent mapping points that each had a voltage of ≤0.5 mV. The left atrium was divided into six segments to describe the location of a LVA as follows: septal, roof, posterior, anterior, inferior, and posterolateral.[12]

In the control group, patients with paroxysmal AF underwent PVI alone without any additional substrate modification (*Figure 1*). In patients

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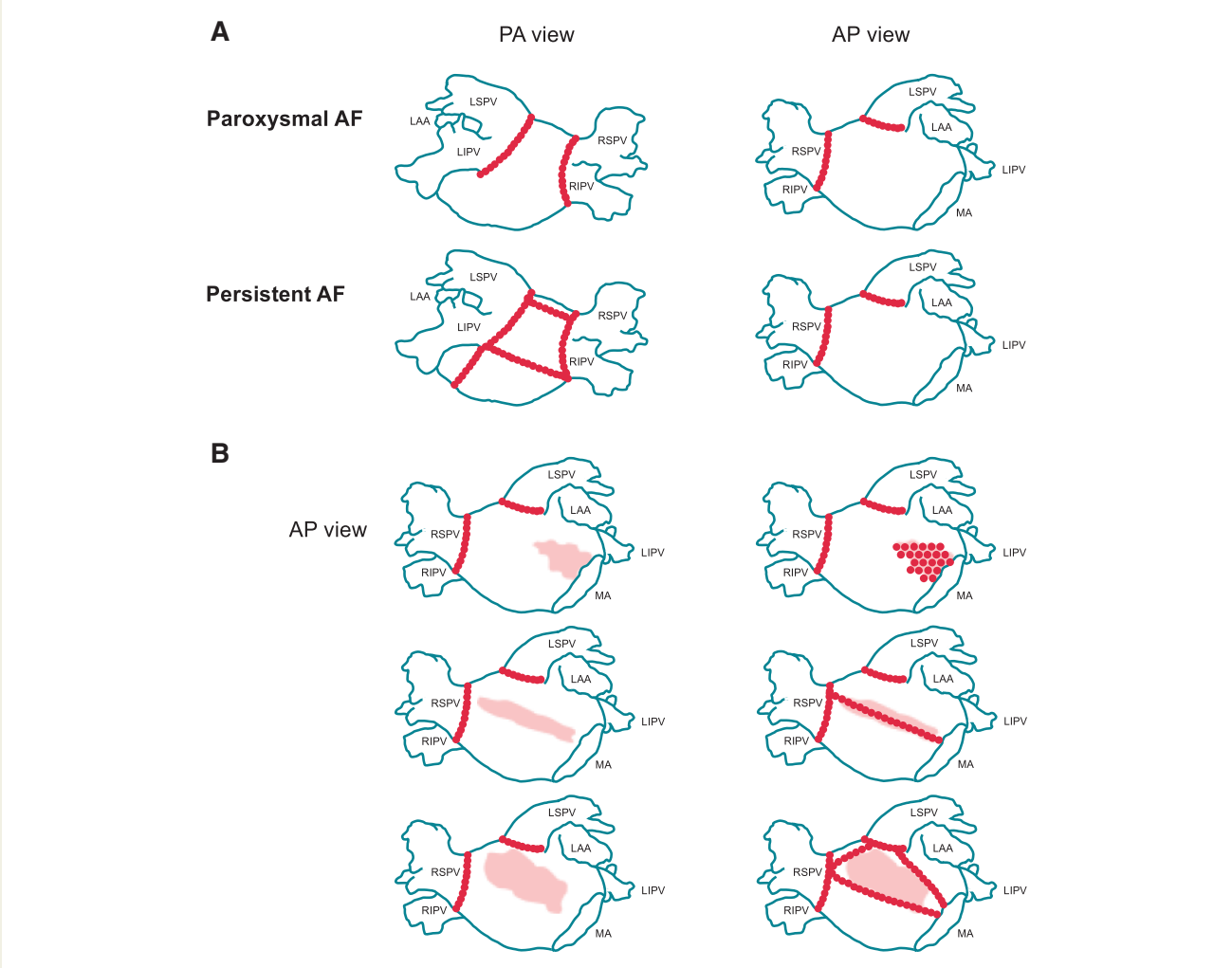

Figure 1 Schematic representation of the strategies for substrate modification in both study groups. Panel (*A*) shows the lesion sets in patients with paroxysmal atrial fibrillation (top) or persistent forms of atrial fibrillation (bottom) in the control group. See text for further information. Panel (*B*) displays three examples of voltage-guided substrate modification. Smaller circumscribed low-voltage areas (pink areas) were approached by regional ablation aiming at tissue homogenization (top). Larger longitudinally oriented low-voltage areas were targeted by linear lesions traversing the low-voltage area and connecting non-excitable tissue (middle). Broader low-voltage areas extending over a wider area were approached by electrical isolation from the remainder myocardium by linear lesions that were connected to non-conducting atrial structures (bottom). AF, atrial fibrillation; AP, anterior–posterior; LAA, left atrial appendage; LIPV, left inferior pulmonary vein; LSPV, left superior pulmonary vein; MA, mitral annulus; PA, posterior–anterior; RIPV, right inferior pulmonary vein; RSPV, right superior pulmonary vein.

with persistent forms of AF, adjunctive linear lesions were applied at the LA roof and bottom connecting the circular lesions around the PVs to isolate the LA posterior wall ('posterior box lesion') and between the left inferior PV and the mitral annulus ('posterior mitral isthmus line'). Isolation of the LA posterior wall was confirmed by entrance and exit block during mapping and pacing at multiple sites inside the 'box'. Bidirectional conduction block across the mitral isthmus line was verified by (i) the presence of widely separated double potentials, (ii) a reduction or a loss of local signals, (iii) the failure to capture the left atrium during pacing along the line, and (iv) differential pacing manoeuvres.[12] If necessary, epicardial ablation via the coronary sinus was applied to achieve complete conduction block across the mitral isthmus.

In the LVA ablation group, PVI was followed by additional substrate modification only if LVAs were detected during EVM irrespective of AF type. The ablation techniques for voltage-guided substrate modification have been detailed previously.[12] In brief, substrate modification consisted of regional LVA homogenization, placement of strategic linear lesions connecting non-excitable tissue, or electrical isolation of the LVA depending on the location, the shape, and the extent of a LVA (*Figures 1* and *2*). Endpoints for LVA homogenization included defractionation, significant reduction in local potentials, and loss of capture during high-output stimulation in that area. Conduction block across linear lesions and electrical isolation of circumscribed areas were confirmed as described for control group patients.

In both study groups, atrial burst pacing was performed to test for inducibility after completion of the intended lesion set.[12] Inducible organized ATs were mapped and ablated with termination and non-inducibility serving as endpoints. If AF or an instable AT was induced, electrical cardioversion was performed without any additional ablation. Pharmacological provocative manoeuvres to unmask dormant PV

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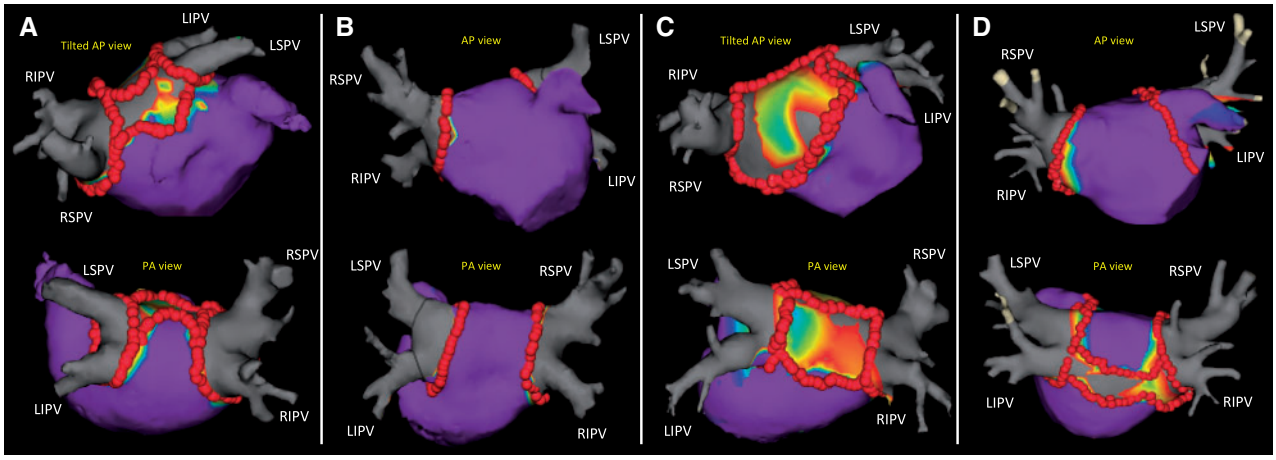

Figure 2 Examples of voltage-guided substrate modification. This figure shows four examples of voltage-guided substrate modification in patients with paroxysmal (*A*) and persistent atrial fibrillation (*B–D*). Purple areas indicate normal voltages, grey areas indicate scar or absent electrical activity, and the remaining colours represent low-voltage areas. Red dots denote ablation points. AP, anterior–posterior; LIPV, left inferior pulmonary vein; LSPV, left superior pulmonary vein; PA, posterior–anterior; RIPV, right inferior pulmonary vein; RSPV, right superior pulmonary vein.

conduction or to identify extra-PV triggers were not applied. Ablation of the cavotricuspid isthmus was performed in patients with documented or induced typical AFLU.

Post-ablation management and follow-up

Class I/III AAD treatment could be reinitiated during the BP in the event of early arrhythmia recurrence. The study protocol, however, required discontinuation of Class I/III AADs before the end of the BP unless indicated by the patients' clinical course. Oral anticoagulation was continued for a minimum of 3 months after ablation, and after 3 months, continuation of oral anticoagulation was determined by the individual stroke risk and the net clinical benefit.[1,2] A post-interventional endoscopy was performed to rule out thermal injury to the oesophagus if the intra-oesophageal temperature exceeded 41 °C and/or if RF energy was delivered near the oesophagus. Serial 7 day Holter recordings were obtained at 3, 6, and 12 months to assess rhythm outcome. Additionally, any post-blanking arrhythmia episode documented outside the Holter recording periods was included in the primary outcome analysis. Patients were strongly encouraged to obtain an ECG if symptoms suggestive of arrhythmia recurrence occurred. All electrograms were analysed by clinicians who were blinded to treatment allocation.

Statistical analysis

The time to the first arrhythmia episode was estimated with the Kaplan–Meier method. Time-to-event analyses were performed using the log-rank test. Outcome analyses were based on a modified intention-to-treat analysis including all patients who underwent ablation and entered the post-blanking follow-up period. For comparisons among study groups, the independent sample Student's *t*-test or the Mann–Whitney *U* test was used for continuous variables as appropriate, and the Fisher's exact test was used for categorical variables. A two-sided *P*-value <0.05 was considered statistically significant. Endocardial voltage mapping was strongly encouraged in the control group to enable comparison between study groups. Since EVM was not an essential part of the ablation procedure in the control group, procedure duration and fluoroscopy time were calculated as follows for this group: Corrected time = total time - time needed for EVM. All statistical analyses were performed with the use of SPSS software (IBM Corp., Armonk, NY, USA).

Results

Patients

From October 2010 to November 2011, 124 patients were randomly assigned to conventional or voltage-guided substrate modification (*Figure 3*). There was a trend towards a higher prevalence of persistent AF in the LVA ablation group (*Table 1*). None of the patients had long-standing persistent AF. In total, 59 patients in each group entered the post-blanking follow-up period. Significantly more patients in the control group received Class I/III AADs after the BP [17/59 (29%) vs. 6/59 (10%), $P = 0.02$]. None of the patients in either of the study group received temporary amiodarone therapy for rhythm control during the BP.

Procedural data

Low-voltage areas were found in 35 /119 patients (29%) in the total study population with a significant difference between patients with persistent and paroxysmal AF [24/59 (41%) vs. 11/60 (18%), $P = 0.009$]. There was a trend towards a higher prevalence of LVAs in the control group when patients with paroxysmal AF were compared (*Table 2*). No between-group difference could be found with respect to any other baseline characteristic including pre-ablation AAD use [32/37 (87%) vs. 25/26 (96%), $P = 0.39$] in the subgroup of patients with paroxysmal AF. Low-voltage areas were predominantly located at the LA posterior wall, at the LA roof, and at the inter-atrial septum. Procedure duration and fluoroscopy time were similar in both groups. Ablation time was significantly shorter in the LVA ablation group only in patients with persistent AF. In the control group, 22/25 patients (88%) with persistent AF underwent PVI plus empirical linear ablation as per protocol. Bidirectional conduction

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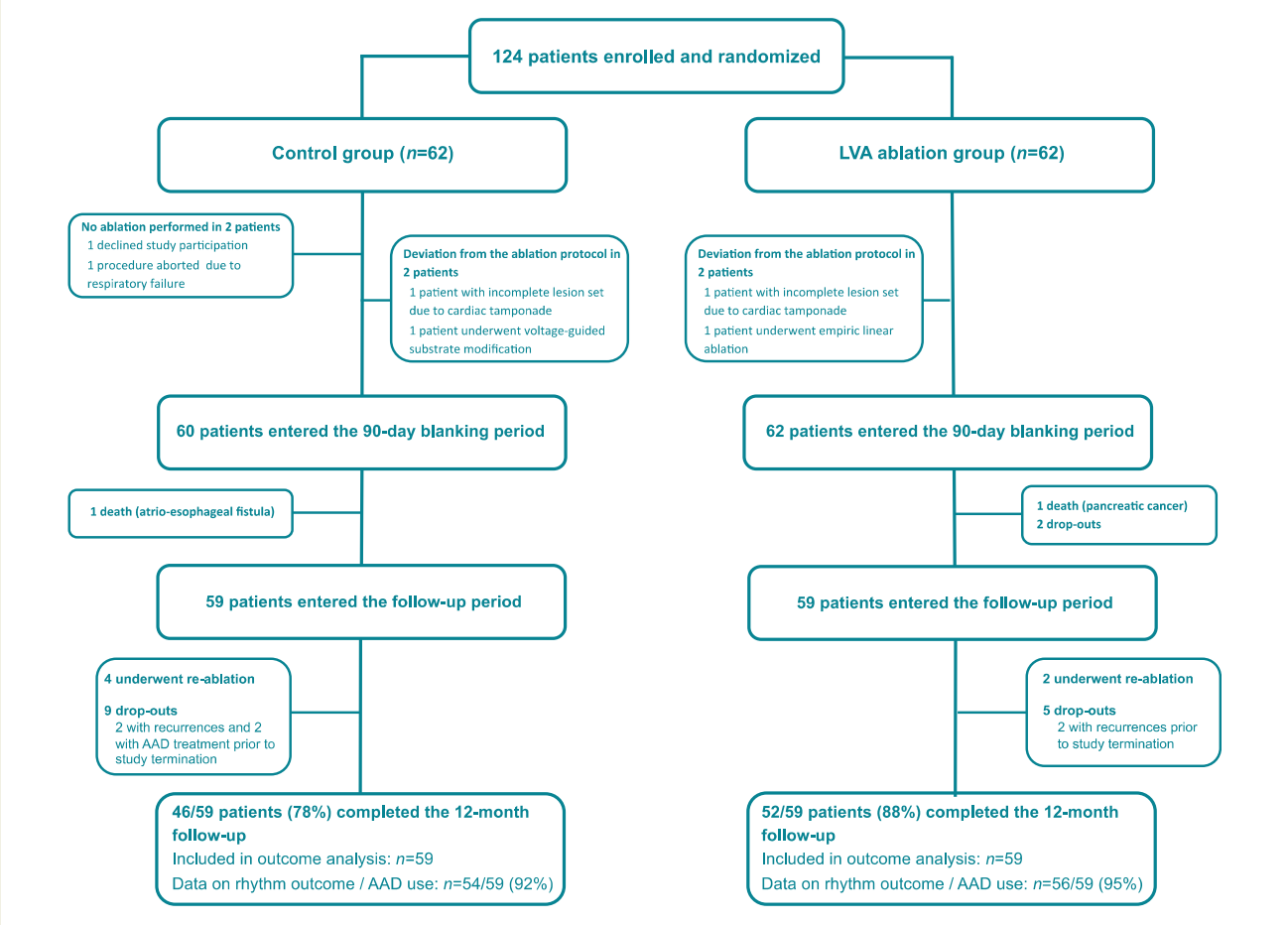

Figure 3 Patient flow diagram. In the control group, 2/62 patients (3%) did not undergo catheter ablation. One patient with persistent atrial fibrillation declined study participation before the procedure. In another patient with paroxysmal atrial fibrillation, the procedure had to be aborted due to severe respiratory failure at the beginning of the procedure before radiofrequency ablation was initiated. In 58/60 control group patients (97%) who underwent ablation the final lesion set complied with the study protocol [in 36/36 patients (100%) with paroxysmal AF, and in 22/24 patients (92%) with persistent AF]. In one patient with persistent atrial fibrillation, isolation of the right inferior pulmonary vein and linear ablation could not be accomplished due to the occurrence of a cardiac tamponade. The second protocol deviation was observed in a patient with persistent atrial fibrillation who received voltage-guided substrate modification instead of linear ablation. In the LVA ablation group, 60/62 patients (97%) underwent ablation treatment as per protocol [15/16 patients (94%) with low-voltage areas, and 45/46 patients (98%) without low-voltage areas]. In one patient with paroxysmal atrial fibrillation and low-voltage areas, voltage-guided substrate modification could not be performed after pulmonary vein isolation due to the occurrence of a cardiac tamponade. Another patient with persistent atrial fibrillation and without low-voltage areas received additional empirical linear ablation which constituted a protocol violation. AAD, antiarrhythmic drug; LVA, low-voltage area.

block across the posterior mitral isthmus and complete isolation of the LA posterior wall could be achieved in 16/22 patients (73%) and in 19/22 patients (86%), respectively.

Efficacy endpoints

After a mean follow-up of 12 ± 3 months, 25/59 patients (42%) in the control group and 40/59 patients (68%) in the LVA ablation group were free from any arrhythmia recurrence off AADs after a single ablation procedure (unadjusted log rank $P = 0.003$) (*Table 3*, *Figure 4*). This difference in the primary outcome was maintained when subgroups with paroxysmal and persistent AF were compared (*Table 3*). Arrhythmia-free survival on or off AADs was observed in 33/59 patients (56%) in the control group and in 41/59 patients (70%) in the LVA ablation group (adjusted log rank $P = 0.10$). During the 7 day Holter at 6 and 12 months, significantly more patients in the LVA ablation group were free from arrhythmia recurrence off AADs [43/55 (78%) vs. 31/54 (57%), $P = 0.03$; and 44/50 (88%) vs. 29/46 (63%), $P = 0.008$] (*Table 3*, Supplementary material online, *Figure S2*). At 12 months, significantly more patients in the LVA ablation group were free from arrhythmia recurrence regardless of AAD use [45/50 (90%) vs. 33/46 (72%), $P = 0.04$]. Among patients with arrhythmia recurrence, there was no significant between-group difference in the rate of organized ATs.

In a subgroup analysis, the presence of LVAs was associated with a lower rate of freedom from arrhythmia recurrence off AADs after a single procedure in each of the study groups, albeit this difference did not reach the level of statistical significance [control group: 19/37 patients (51%) without LVAs vs. 5/19 patients (26%) with LVAs,

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Table 1 Baseline characteristics

	Control group ($n = 62$)	LVA ablation group ($n = 62$)	*P*-value
Patient characteristics			
Age (years)	63 ± 9	62 ± 10	0.47
Male gender	41 (66)	36 (58)	0.46
Persistent atrial fibrillation	25 (40)	36 (58)	0.07
Atrial fibrillation history (months)	60 (36–111)	54 (24–87)	0.21
Hypertension	50 (81)	49 (79)	1.00
Structural heart disease	12 (19)	12 (19)	1.00
CHA_2DS_2-VASc score	2 (1–3)	2 (1–3)	0.92
Implantable cardiac device[a]	6 (10)	9 (15)	0.58
Left ventricular ejection fraction (%)	61 ± 7	59 ± 9	0.17
Left atrial diameter (mm)	42 ± 6	43 ± 6	0.37
Prior Class I/III antiarrhythmic drugs	56 (90)	50 (81)	0.20
Amiodarone use <3 months before ablation	7 (11)	8 (13)	1.00
Medication at baseline			
Class I or III antiarrhythmic drugs	26 (42)	30 (48)	0.59
β-Blocker	53 (86)	55 (89)	0.79
Calcium-channel blocker	2 (3)	4 (7)	0.68
Cardiac glycoside	9 (15)	8 (13)	1.00
Vitamin K antagonist	43 (69)	41 (66)	0.85
Acetylsalicylic acid	17 (27)	13 (21)	0.53
No antithrombotic therapy	3 (5)	8 (13)	0.21

Values are represented as mean ± standard deviation, median (interquartile range), or *n* (%).
[a]Including all cardiac devices capable of supraventricular arrhythmia detection and storage.

log-rank $P = 0.11$; LVA ablation group: 32/43 patients (74%) without LVAs vs. 8/16 patients (50%) with LVAs, log-rank $P = 0.06$].

Safety endpoints

Major complications were observed in 11/123 patients (9%) with no significant difference among groups (*Table 4*). A fatal atrio-oesophageal fistula occurred in a 72-year-old female patient with paroxysmal AF in the control group who underwent PVI alone as per protocol. A post-interventional endoscopy was performed in 35/60 patients (58%) in the control group and in 36/62 patients (58%) in the LVA ablation group that revealed thermal lesions in 6/35 patients (17%) and in 3/36 patients (8%), respectively ($P = 0.31$).

Discussion

Main results

In this single-centre randomized trial, individualized substrate modification guided by EVM was associated with a significantly higher drug-unassisted arrhythmia-free survival rate after a single AF ablation procedure compared with a conventional approach applying empirical linear lesions according to AF type. This difference in the primary endpoint was maintained in the subsets of patients with paroxysmal and persistent AF. In control group patients with paroxysmal AF, the low arrhythmia-free survival may be at least partially explained by the (numerically) higher prevalence of LVAs that were left untreated as per protocol. In patients with persistent AF, however, voltage-guided ablation was associated with a lower incidence of arrhythmia recurrence compared to empirical linear ablation with no relevant between-group difference found in the prevalence of LVAs or other baseline characteristics. Freedom from arrhythmia recurrence in control group patients with persistent AF was comparable to that reported in the 'linear ablation' arm in the large-scale randomized STAR AF II study.[15] The success rate in the control group of this study is lower than previously reported by our group in a similar patient population undergoing AF ablation.[16] This discrepancy may be explained by differences in the design of the studies (prospective vs. retrospective), differences in AAD use, different definitions of paroxysmal and persistent AF, and the unexpectedly low rate of freedom from arrhythmia recurrence in patients with paroxysmal AF in the control group of this study.

There was no difference in the rate of peri-interventional complications between groups. The fatal atrio-oesophageal fistula was not associated with any form of substrate modification strategy since the patient had paroxysmal AF without LVAs and would thus undergo PVI alone regardless of treatment allocation. As reported previously, LVAs were more frequently found in patients with persistent AF compared to those with paroxysmal AF.[12] Remarkably, approximately one-fifth of patients with paroxysmal AF exhibited LVAs, whereas more than half of the patients with persistent AF had normal voltages.

Clinical role of low-voltage areas

Left atrial structural changes, particularly fibrosis, play a key role in the pathogenesis of AF.[5–8] Fibrosis is associated with impaired cellular

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Table 2 Procedural data

	Control group ($n = 60$)[a]	LVA ablation group ($n = 62$)	*P*-value
Procedure duration[b] (min)	164 ± 48	161 ± 43	0.68
Paroxysmal AF	150 ± 47	154 ± 41	0.75
Persistent AF	185 ± 40	166 ± 44	0.09
Fluoroscopy time[b] (min)	22.7 (17.9–34.4)	24.5 (20.0–31.6)	0.40
Paroxysmal AF	19.3 (16.9–27.2)	22.0 (17.7–34.6)	0.23
Persistent AF	34.9 (21.5–41.2)	24.7 (21.5–30.8)	0.09
Ablation time (min)	41.4 (28.5–51.3)	37.3 (25.9–55.0)	0.43
Paroxysmal AF	36.3 (24.0–47.8)	38.9 (30.9–62.4)	0.14
Persistent AF	47.9 (43.0–56.6)	33.9 (24.5–47.4)	0.002
Dose–area product (Gy·cm^2)	91.6 (53.8–156.3)	88.0 (52.0–136.3)	0.65
Paroxysmal AF	80.5 (52.3–108.3)	72.4 (37.8–106.0)	0.24
Persistent AF	105.6 (58.8–184.5)	105.3 (61.0–177.3)	0.68
Voltage map created	57 (95)	62 (100)	0.12
Time needed for EVM (min)	12.0 (10.0–15.0)	12.5 (9.0–19.3)	0.59
Fluoroscopy time for EVM (min)	2.2 (1.5–3.0)	2.3 (1.3–3.7)	0.75
PVI achieved	59 (98)	61 (98)	0.98
Treatment as per protocol[c]	58 (97)	60 (97)	1.00
LVAs present	19 (33)	16 (26)	0.42
Paroxysmal AF	9 (27)	2 (8)	0.09
Persistent AF	10 (44)	14 (39)	0.79
Left atrial segments with LVAs	2.8 ± 1.9	2.3 ± 1.7	0.45
Location of LVAs			
Posterior	9 (47)	7 (44)	1.00
Septal	7 (37)	6 (38)	1.00
Roof	9 (47)	6 (38)	0.73
Posterolateral	1 (5)	1 (6)	1.00
Anterior	1 (5)	2 (13)	0.58
Inferior	1 (5)	2 (13)	0.58
Ubiquitous[d]	4 (21)	2 (13)	0.67
Cavotricuspid isthmus ablation	9 (15)	10 (16)	1.00

Values are represented as mean ± standard deviation, median (interquartile range), or *n* (%).
AF, atrial fibrillation; EVM, endocardial voltage mapping; LVA, low-voltage area; PVI, pulmonary vein isolation.
[a]Only patients in whom at least one radiofrequency pulse was applied were included in the analysis of procedural data (see also *Figure 3*).
[b]In the control group, procedure duration and fluoroscopy time were calculated as follows: Corrected time = total time - time needed for voltage mapping (see text for further information).
[c]See *Figure 3* for further information.
[d]Presence of LVAs in >4 left atrial segments.

coupling, heterogeneity of intra-atrial conduction, and dispersion of refractory periods, which provide the substrate for re-entry capable of sustaining AF.[6,14,17] Structural changes of the atrial myocardium can be non-invasively detected and quantified by DE MRI and manifest in one or more LA segments with a high inter-individual variability of regional distribution.[5,8,9,12] Since LVAs correspond well to structural alterations identified by DE MRI, electroanatomic mapping can be used for individualized substrate description.[9] In the absence of any clear evidence, it may be assumed that fibrosis predominantly forms the histopathological correlate of LVAs although other tissue alterations (e.g. fatty infiltration and amyloidosis) may contribute to reduced voltages.[14] Clinical trials consistently reported that both the presence and the degree of structural alterations were highly predictive of treatment failure after AF ablation.[5,7,8] These findings form the rationale for voltage-guided substrate modification.

Our current findings are in line with previous reports. We have recently studied the impact of voltage-guided substrate modification on rhythm outcome in a retrospective observational study including 178 ablation-naive patients with paroxysmal or persistent AF.[12] After 12 months of follow-up, similar rates of arrhythmia-free survival were observed in patients with LVAs undergoing PVI plus individualized substrate modification and in patients without LVAs undergoing PVI alone [33/47 (70%) vs. 81/131 (62%), $P = 0.3$] suggesting that targeting LVAs by RF ablation may counteract their arrhythmogenicity. Similar results were reported in a study by Yamaguchi *et al.*[18] including 101 patients with exclusively persistent AF. After 18 ± 7 months, similar rates of single-procedure freedom from arrhythmia recurrence were found in patients with LVAs undergoing PVI plus voltage-guided ablation and in patients without LVAs undergoing PVI alone [28/39 (72%) vs. 49/62 (79%), $P = 0.4$]. Furthermore, voltage-guided substrate modification

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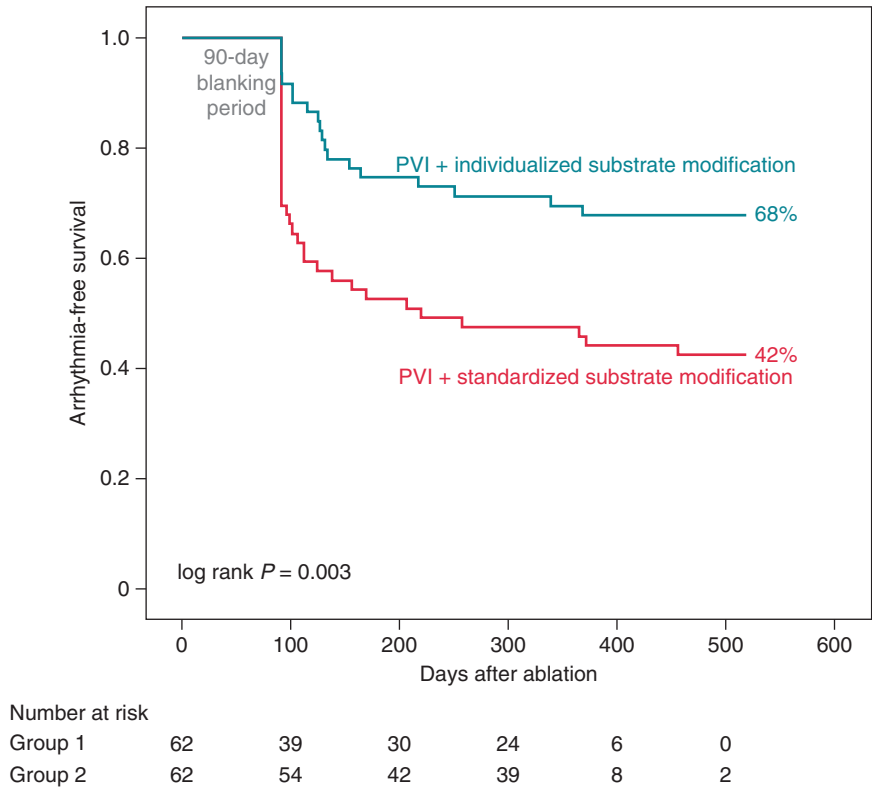

Figure 4 Primary efficacy outcome. The Kaplan–Meier estimates of single-procedure freedom from any atrial arrhythmia >30 s off antiarrhythmic medication. PVI, pulmonary vein isolation.

was associated with a significantly higher arrhythmia-free survival rate compared to a step-wise approach consisting of linear ablation and electrogram-based ablation.[19,20] In a randomized study by Wang *et al.*[20] including 124 patients with long-standing persistent AF, 42/64 patients (66%) in the voltage-guided ablation group and 27/60 patients (45%) in the step-wise ablation group were free from arrhythmia recurrence on or off AADs after a single procedure at 12 months of follow-up ($P = 0.04$). To the best of our knowledge, this study is the first randomized trial to investigate the role of voltage-based ablation in patients with paroxysmal or persistent AF who represent the largest group among patients undergoing catheter ablation for AF.[21]

Comparison with other substrate modification strategies

There is widespread consensus that PVI should be the minimum lesion set for both paroxysmal and persistent AF.[1,2] Adjunctive substrate modification is frequently applied in persistent forms of AF in an effort to improve ablation success. This strategy is based on the observation that mechanisms outside the PV antra may be operative in arrhythmia maintenance in persistent forms of AF. Different strategies have been proposed to target substrate-related mechanisms such as ablation of complex fractionated atrial electrograms (CFAEs) or anatomically defined empirical linear ablation.[3,4] However, available data concerning the benefit of such strategies are conflicting.[3,4,15] In the STAR AF II trial, additional substrate modification (i.e. CFAE ablation or linear ablation) did not translate into an improved outcome when compared with PVI alone in patients with persistent AF.[15] These data suggest that current strategies for substrate modification fail to consistently address patient-specific mechanisms of AF perpetuation and that elimination of PV arrhythmogenicity by antral PVI may be sufficient in a considerable subset of patients with persistent AF.

There is a great overlap in the prevalence of LA structural alterations when patients with paroxysmal and persistent AF were compared.[12] This ill-defined association between the clinical presentation and the stage of atrial disease and the highly variable inter-individual distribution of LVAs emphasize the need for a more patient-specific substrate description.[6,12] In this study, a considerable number of patients with paroxysmal AF would be undertreated by PVI alone. On the contrary, the majority of patients with persistent AF had no LVAs and would thus be overtreated by empirical linear ablation which may be technically challenging and potentially proarrhythmogenic due to gap-related re-entrant tachycardias.[22] Moreover, the standardized linear lesion set that was applied in our study consisting of a 'box lesion' and a mitral isthmus line would miss atrial substrate in a considerable proportion of patients. Thus, empirical linear ablation may be associated with at least partial undertreatment despite extensive ablation.

Limitations

Our study has several limitations that need to be discussed. First, outcome data in the non-pre-specified subgroups of patients with

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Table 3 Efficacy outcomes

Freedom from any arrhythmia recurrence	Control group	LVA ablation group	*P*-value
off AADs	25/59 (42)	40/59 (68)	0.003[a]
Paroxysmal AF	16/35 (46)	18/25 (72)	0.04[a]
Persistent AF	9/24 (38)	22/34 (65)	0.03[a]
On or off AADs	33/59 (56)	41/59 (70)	0.10[b]
Paroxysmal AF	21/35 (60)	19/25 (76)	0.23
Persistent AF	12/24 (50)	22/34 (65)	0.26
At 6 months off AADs[c]	31/54 (57)	43/55 (78)	0.03
Paroxysmal AF	20/32 (63)	19/24 (79)	0.24
Persistent AF	11/22 (50)	24/31 (77)	0.046
At 6 months on or off AADs[c]	42/54 (78)	45/55 (82)	0.64
Paroxysmal AF	27/32 (84)	20/24 (83)	1.00
Persistent AF	15/22 (68)	25/31 (81)	0.35
At 12 months off AADs[c]	29/46 (63)	44/50 (88)	0.008
Paroxysmal AF	20/28 (71)	17/21 (81)	0.44
Persistent AF	09/18 (50)	27/29 (93)	0.001
At 12 months on or off AADs[c]	33/46 (72)	45/50 (90)	0.04
Paroxysmal AF	23/28 (82)	18/21 (86)	1.00
Persistent AF	10/18 (56)	27/29 (93)	0.004
Rate of organized atrial tachycardias	13/26 (50)	10/18 (56)	0.77
Paroxysmal AF	6/14 (43)	4/6 (67)	0.63
Persistent AF	7/12 (58)	6/12 (50)	1.00

Values are *n* (%).
AAD, antiarrhythmic drug; AF, atrial fibrillation.
[a]Unadjusted log rank *P*.
[b]Log-rank *P* adjusted for type of AF.
[c]Only arrhythmia recurrences detected during the 7 day Holter ECG recording periods at 6 and 12 months were included.

Table 4 Procedure-related adverse events

Type of adverse event	Total (*n* = 123)	Control group (*n* = 61)	LVA ablation group (*n* = 62)	*P*-value
Cardiac tamponade	4 (3)	2 (3)	2 (3)	1.00
Stroke or transient ischaemic attack	1 (1)	1 (2)	0 (0)	0.50
Pericarditis	1 (1)	0 (0)	1 (2)	1.00
Periprocedural respiratory failure	1 (1)	1 (2)	0 (0)	0.50
Mild groin haematoma	6 (5)	3 (5)	3 (5)	1.00
Major vascular access site complications[a]	4 (3)	2 (3)	2 (3)	1.00
Atrio-oesophageal fistula	1 (1)	1 (2)	0 (0)	0.50
Endoscopy performed[b]	71 (58)	35 (58)	36 (58)	1.00
Thermal lesions	9 (13)	6 (17)	3 (8)	0.31
Major complications[c]	11 (9)	7 (11)	4 (6)	0.36

Values are represented as *n* (%). Only events that occurred during or after the first ablation procedure were included.
[a]Defined as those requiring intervention, or prolonging or requiring hospitalization.
[b]Only patients who underwent ablation were included in the analysis.
[c]Defined as those that result in permanent injury or death, require intervention for treatment, or prolong or require hospitalization.[1]

paroxysmal and persistent AF have to be interpreted with caution due to small sample sizes. Second, randomization was not stratified with regard to the type of AF or the presence of LVAs, which would have prevented the statistically non-significant, but potentially relevant, between-group differences in these characteristics that have been observed in this study despite randomization. It may be assumed that the higher prevalence of structural alterations in control group patients with paroxysmal AF is causally linked to the worse outcome in the entire control group. This association, however, reinforces the concept that LVAs may serve as potential ablation

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targets for patient-specific substrate modification. The difference in the type of AF at baseline favoured the control group, because more patients with persistent AF were found in the LVA ablation group. Third, generalizability of reported data is limited, as they represent a single-centre experience with site-specific use of technologies and techniques. Fourth, the impact of either ablation strategy on quality of life has not been addressed. Short-lasting arrhythmia episodes that were considered a treatment failure may not have a meaningful clinical impact. This study, however, was specifically designed to evaluate the efficacy of different ablation strategies using the primary endpoint that is proposed by expert consensus.[1] Fifth, only 29% of the entire study population had LVAs on EVM. As a consequence, the majority of patients with paroxysmal AF in either study group received an identical ablation (i.e. PVI alone) as per protocol. The primary goal of this investigation, however, was to evaluate the efficacy and the safety of two entirely different approaches for substrate modification in a representative 'real-world' patient cohort. Exclusion of patients without LVAs would have produced significant selection bias to the disadvantage of the control group, because patients with LVAs represent a specific subgroup among patients undergoing AF ablation with a more advanced atrial disease. Sixth, the between-group difference in AAD use after the BP has substantially contributed to the difference in the primary efficacy endpoint. Post-interventional prescription of AADs, however, reflected clinical decision-making in an individual patient and not a protocol violation by the investigators. Seventh, the study was underpowered to detect the effect of LVAs on ablation outcome irrespective of the ablation strategy. And, finally, blinding of the study participants with regard to the allocated treatment could not be fully ensured as patients had access to the procedural report.

Conclusions

Voltage-guided ablation provides a novel technique for patient-specific substrate modification with a clear definition of ablation target sites. This study found that individually tailored substrate modification was associated with a significantly higher single-procedure arrhythmia-free survival rate off AADs compared with a conventional approach applying empirical substrate modification only in patients with persistent AF. Future well-designed clinical trials are needed to investigate the role of voltage-guided substrate modification, including a comparison of PVI alone with PVI plus voltage-guided ablation. Additionally, future research should focus on methodological aspects of voltage-guided ablation such as the optimal number of mapping points acquired during EVM and individualized voltage cut-offs for the definition of ablation targets.

Supplementary material

Supplementary material is available at *Europace* online.

Conflict of interest: none declared.

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